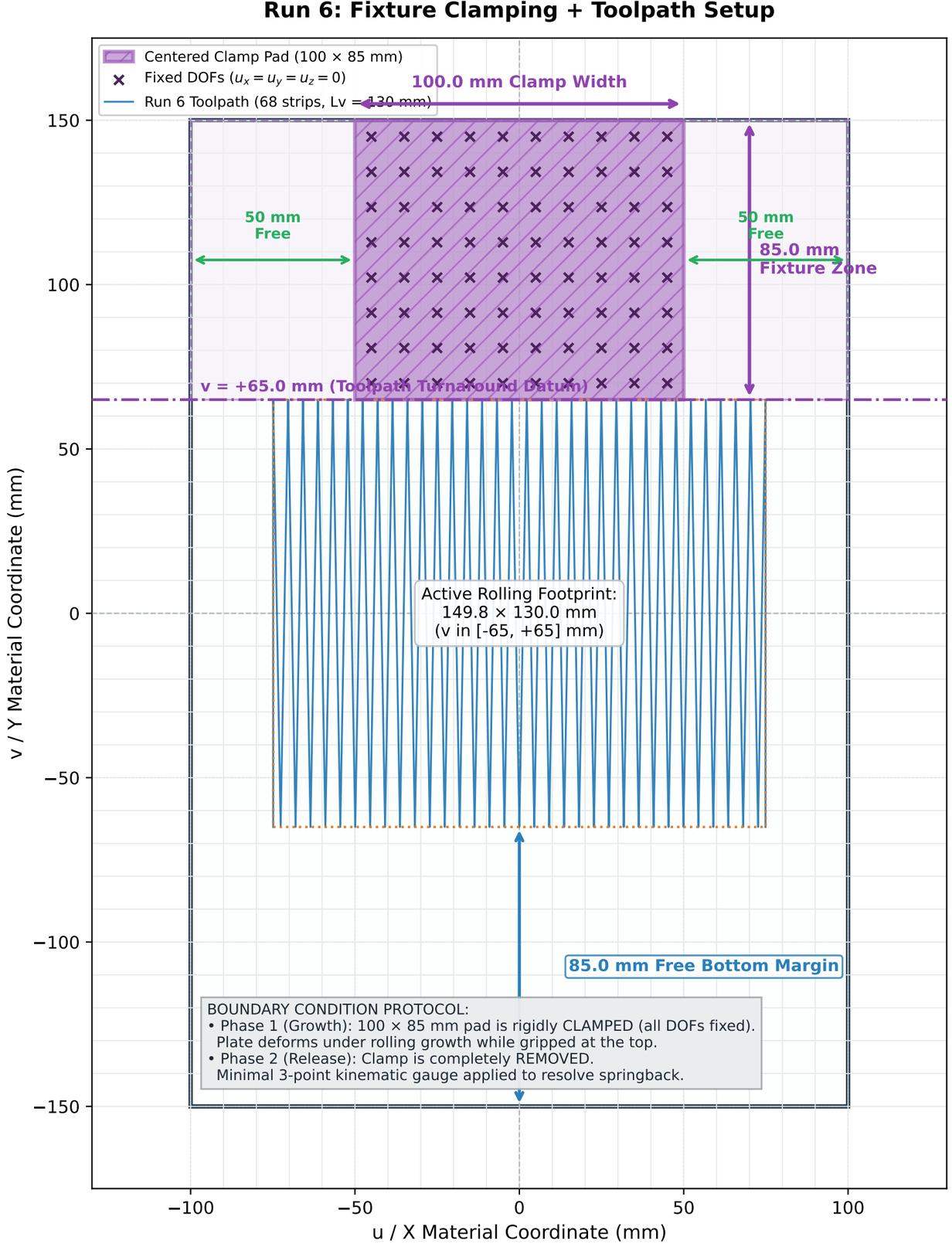
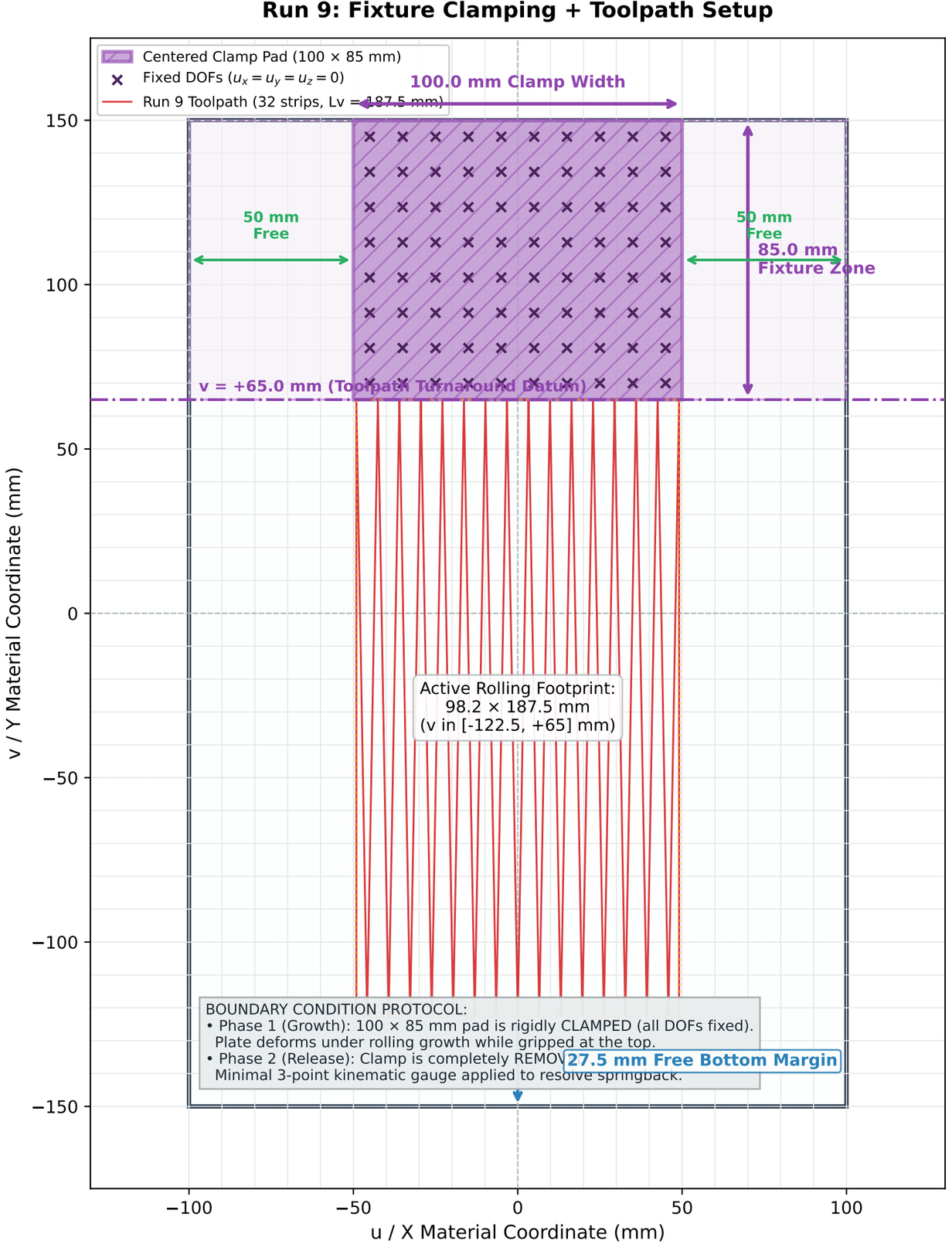

Run 6: Fixture Clamping + Toolpath Setup
Centered Clamp Pad (100 × 85 mm)
Fixed DOFs ($u_x = u_y = u_z = 0$)
Run 6 Toolpath (68 strips, Lv = 130 mm)
100.0 mm Clamp Width
50 mm Free
50 mm Free
85.0 mm Fixture Zone
v = +65.0 mm (Toolpath Turnaround Datum)
Active Rolling Footprint:
149.8 × 130.0 mm
(v in [-65, +65] mm)
85.0 mm Free Bottom Margin
BOUNDARY CONDITION PROTOCOL:
• Phase 1 (Growth): 100 × 85 mm pad is rigidly CLAMPED (all DOFs fixed).
Plate deforms under rolling growth while gripped at the top.
• Phase 2 (Release): Clamp is completely REMOVED.
Minimal 3-point kinematic gauge applied to resolve springback.
v / Y Material Coordinate (mm)
u / X Material Coordinate (mm)
150
100
50
0
−50
−100
−150
−100
−50
0
50
100


Run 9: Fixture Clamping + Toolpath Setup
Centered Clamp Pad (100 × 85 mm)
Fixed DOFs ($u_x = u_y = u_z = 0$)
Run 9 Toolpath (32 strips, Lv = 187.5 mm)
100.0 mm Clamp Width
50 mm Free
50 mm Free
85.0 mm Fixture Zone
v = +65.0 mm (Toolpath Turnaround Datum)
Active Rolling Footprint:
98.2 × 187.5 mm
(v in [-122.5, +65] mm)
27.5 mm Free Bottom Margin
BOUNDARY CONDITION PROTOCOL:
• Phase 1 (Growth): 100 × 85 mm pad is rigidly CLAMPED (all DOFs fixed).
Plate deforms under rolling growth while gripped at the top.
• Phase 2 (Release): Clamp is completely REMOVED.
Minimal 3-point kinematic gauge applied to resolve springback.
v / Y Material Coordinate (mm)
u / X Material Coordinate (mm)
150
100
50
0
−50
−100
−150
−100
−50
0
50
100